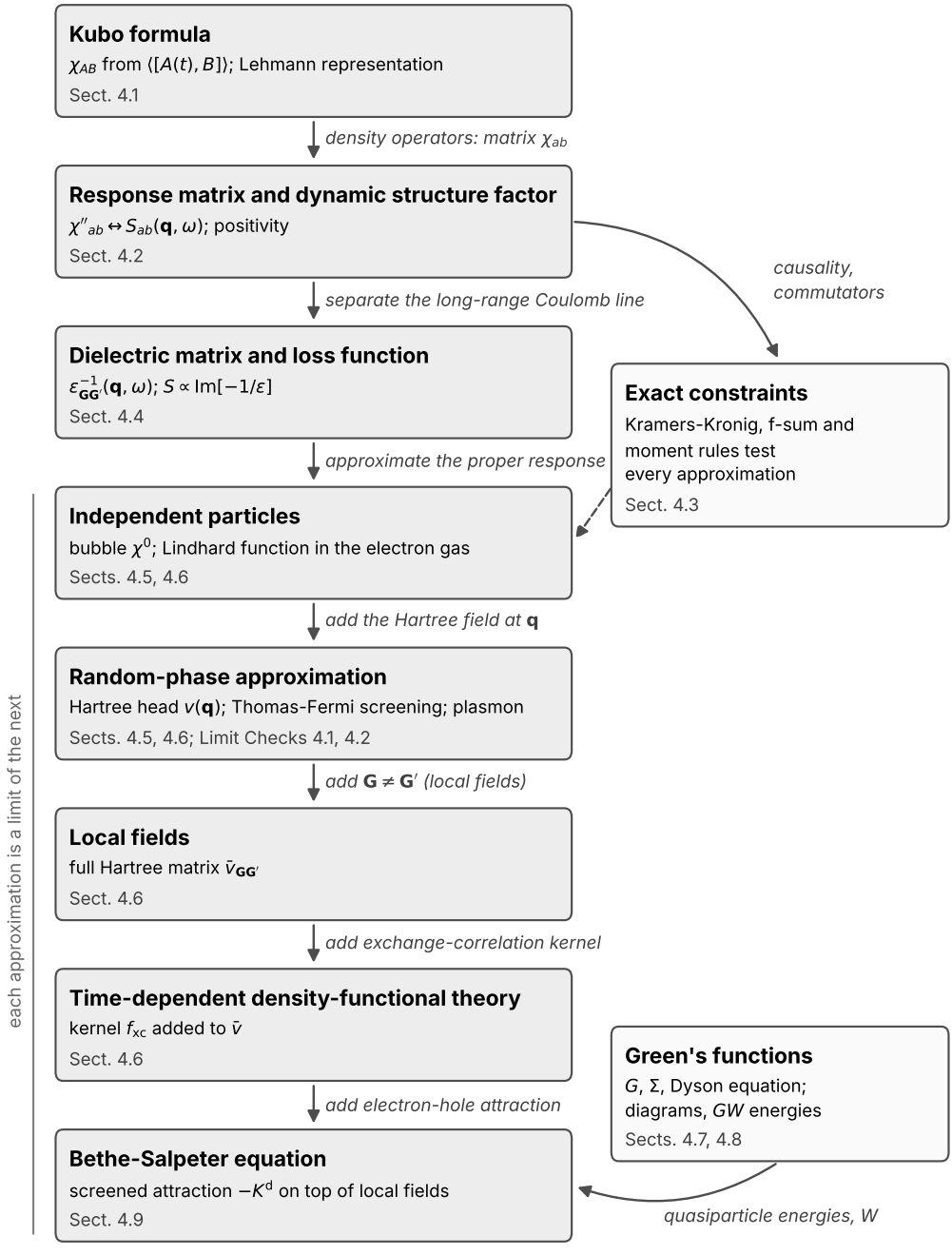
Kubo formula
χ_{AB} from $\langle[A(t), B]\rangle$; Lehmann representation
Sect. 4.1
density operators: matrix χ_{ab}
Response matrix and dynamic structure factor
$\chi''_{ab} \leftrightarrow S_{ab}(\mathbf{q}, \omega)$; positivity
Sect. 4.2
causality, commutators
separate the long-range Coulomb line
Dielectric matrix and loss function
$\varepsilon^{-1}_{\mathbf{G}\mathbf{G}'}(\mathbf{q}, \omega)$; $S \propto \mathrm{Im}[-1/\varepsilon]$
Sect. 4.4
Exact constraints
Kramers-Kronig, f-sum and moment rules test every approximation
Sect. 4.3
approximate the proper response
Independent particles
bubble χ^0; Lindhard function in the electron gas
Sects. 4.5, 4.6
add the Hartree field at $\mathbf{q}$
Random-phase approximation
Hartree head $v(\mathbf{q})$; Thomas-Fermi screening; plasmon
Sects. 4.5, 4.6; Limit Checks 4.1, 4.2
add $\mathbf{G} \neq \mathbf{G}'$ (local fields)
Local fields
full Hartree matrix $\bar{v}_{\mathbf{G}\mathbf{G}'}$
Sect. 4.6
add exchange-correlation kernel
Time-dependent density-functional theory
kernel f_{xc} added to $\bar{v}$
Sect. 4.6
add electron-hole attraction
Bethe-Salpeter equation
screened attraction $-K^d$ on top of local fields
Sect. 4.9
Green's functions
G, Σ, Dyson equation; diagrams, GW energies
Sects. 4.7, 4.8
quasiparticle energies, W
each approximation is a limit of the next